

ELECTRICAL SAFETY

Control electric shock, arc-flash and fire hazards.

PURPOSE AND SCOPE

Review electric-shock, arc-flash, burn, fire and unexpected-energisation hazards associated with electrical equipment, temporary power, cables, sockets, panels and energy sources, and verify the required controls before work begins. Electricity is an invisible hazard. A damaged cable, loose connection, open panel, poor earthing or wet environment can expose a worker to a fatal electric current.

KEY HAZARDS

 ELECTRIC SHOCK	 ARC FLASH	 VOLTAGE	 DAMAGED CABLE
 WRONG TEST	 WET AREA	 OPEN PANEL	 UNAUTHORIZED WORK

CRITICAL VALUES TO VERIFY IN THE FIELD

100% Treat as energized until verified	Test-Measure-Test Absence-of-voltage method	0 Unauthorized energized work
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REMEMBER - SERNEM

- | | | |
|-------------------------|----------------------------|----------------------------------|
| S Stop and think | E Evaluate the risk | R Remove or reduce |
| N Never assume | E Ensure protection | M Monitor and communicate |

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STANDARDS AND FIELD REQUIREMENTS

OSHA 1910.333-335 and NFPA 70E address de-energized work, lockout, absence-of-voltage testing, boundaries and arc-hazard PPE.

OSHA 1910.333(a)

DE-ENERGIZED WORK

De-energize exposed live parts where feasible; do not perform energized work without a documented justification.

OSHA 1910.333(b)

LOCKING AND TAGGING

Disconnect, lock, release stored energy and verify absence of voltage.

OSHA 1910.335

PROTECTIVE EQUIPMENT

Use insulated tools, gloves and face/body protection appropriate to shock and arc hazards.

NFPA 70E

RISK ASSESSMENT

Determine shock and arc-flash risk, boundaries, labeling and PPE appropriate to the incident energy.

MINIMUM EVIDENCE BEFORE START

- | | |
|---|---|
| ☐ Single-line diagram and circuit identification | ☐ Electrical work permit |
| ☐ LOTO record | ☐ Test-instrument suitability record |
| ☐ Shock / arc-risk assessment | ☐ Qualified-electrician record |

SUPERVISOR DISCUSSION POINTS

- Have all cables and electrical tools been inspected?
- Are earthing and residual-current protection suitable?
- Are cables protected from water, edges and vehicles?

ABSENCE-OF-VOLTAGE VERIFICATION



Isolation - Lock - Test-Measure-Test - Boundary - Arc PPE



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CRITICAL CONTROL MEASURES (HIERARCHY OF CONTROLS)

1. DE-ENERGIZE

Inspect equipment, cables, plugs and sockets before use. Remove damaged equipment from service and quarantine it.

2. LOCK AND VERIFY

Verify suitable earthing and residual-current protection. Keep electrical panels closed, locked and accessible. Protect cables from water, sharp edges and vehicle routes.

3. ESTABLISH BOUNDARIES

Do not overload or daisy-chain extension leads. Prioritise de-energised work and apply LOTO. Verify absence of voltage with a suitable tester.

4. USE CORRECT PPE AND TOOLS

Allow only authorised electrical personnel to intervene. Safely isolate power during electrical faults or fires.

CRITICAL WARNING



Do not use electrical equipment showing damage, exposed conductors, burning smell or abnormal heat. Isolate the supply, tag the equipment and report it to an authorised person.

SITE CHECKLIST (BEFORE STARTING WORK)

- | | |
|---|---|
| ☐ Cables intact | ☐ Plugs intact |
| ☐ Sockets suitable | ☐ Earthing available |
| ☐ RCD/GFCI active | ☐ Panels closed |
| ☐ Area dry | ☐ Cables protected |
| ☐ LOTO ready | |



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ATTENDANCE AND APPROVAL

Project / Site: _____ Date: _____

Presented by: _____ Work area: _____

No	Full Name	Company / Role	Signature
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			